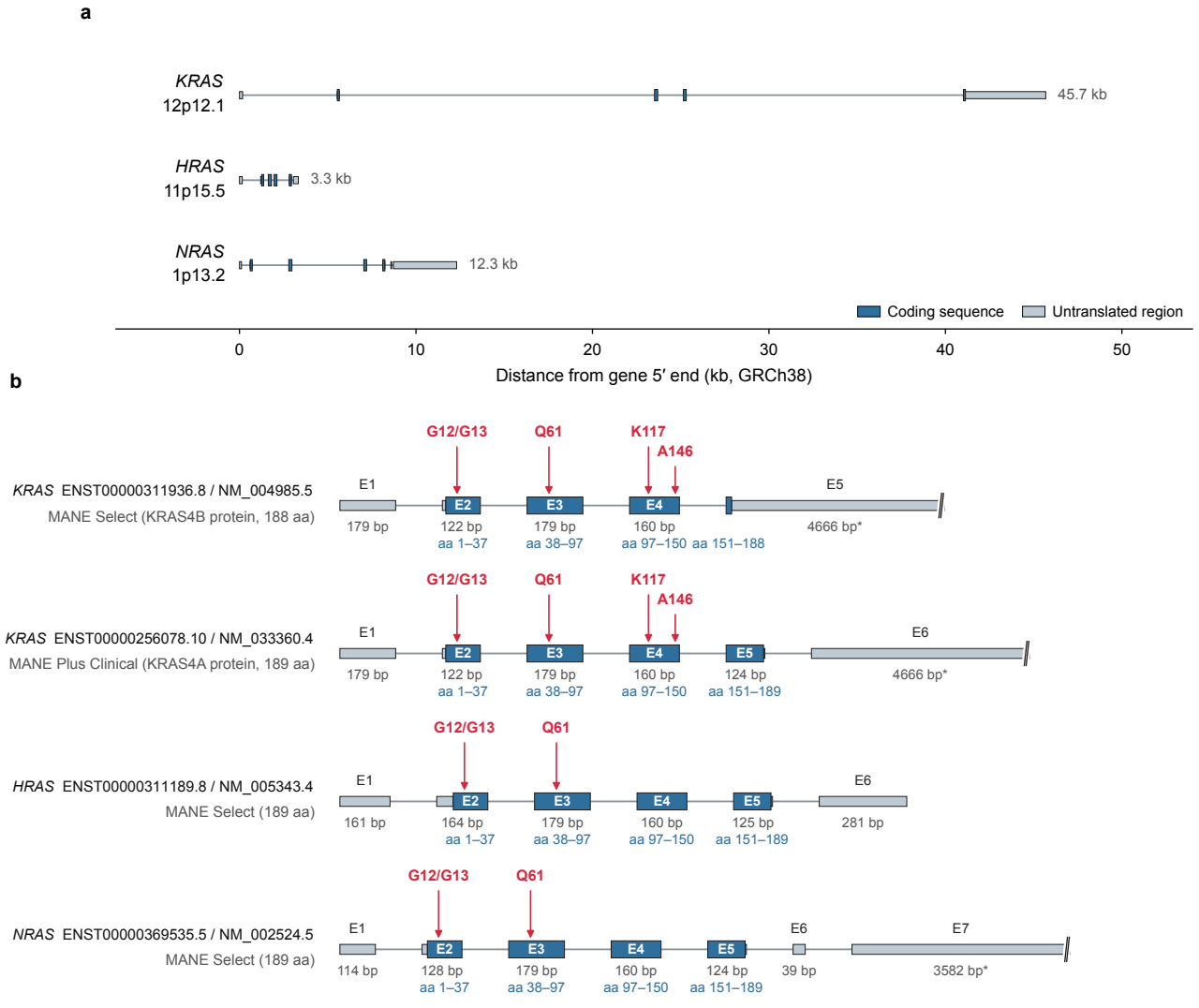
a
KRAS
12p12.1
45.7 kb
HRAS
11p15.5
3.3 kb
NRAS
1p13.2
12.3 kb
Coding sequence
Untranslated region
0
10
20
30
40
50
Distance from gene 5′ end (kb, GRCh38)
b
KRAS ENST00000311936.8 / NM_004985.5
MANE Select (KRAS4B protein, 188 aa)
G12/G13
Q61
K117
A146
E1
E2
E3
E4
E5
179 bp
122 bp
aa 1–37
179 bp
aa 38–97
160 bp
aa 97–150
aa 151–188
4666 bp*
KRAS ENST00000256078.10 / NM_033360.4
MANE Plus Clinical (KRAS4A protein, 189 aa)
E6
124 bp
aa 151–189
HRAS ENST00000311189.8 / NM_005343.4
MANE Select (189 aa)
161 bp
164 bp
281 bp
NRAS ENST00000369535.5 / NM_002524.5
MANE Select (189 aa)
E7
114 bp
128 bp
39 bp
3582 bp*